DocScout-X

Query-Aware Fine-Grained Evidence Retrieval from Long Documents

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Problem Statement: In long documents such as reports, filings and forms, the answer to a question usually sits in one paragraph, table cell or chart buried among many pages. Document- or page-level search returns whole files or pages, flat-text pipelines lose the layout and table structure, and a generated answer without a pointer to its evidence cannot be checked.

Objective: Build a query-aware retrieval system that decides whether a question is textual, numerical or visual, retrieves the exact evidence component (paragraph, table and its matching cells, or figure) inside the document, explains why each item was retrieved (page, box, scores, section path), answers only from that evidence with citations, and improves the ranking within a session from the user's relevance feedback.

Dataset Used: MP-DocVQA (Hugging Face lmms-lab/MP-DocVQA, validation split): 49 multi-page documents, 250 pages, 293 questions, parsed into 2,976 components; split by document into DEV (10 documents, 76 questions, used for all tuning) and TEST (39 documents, 217 questions). Relevance labels: the answer page and every component that contains a gold answer (94.9% of questions covered). Demo document: US Census report P60-281 (31 pages).

Methods Used / Algo Used: Docling layout parsing with TableFormer and RapidOCR into typed components, each serialized to text and linked in a document structure graph (networkx). Three index branches (text, table, figure) with BM25 and BGE-base-en-v1.5 dense retrieval (FAISS), plus CLIP ViT-B/32 image embeddings for figures. A query-type classifier (rules, LLM when unsure) sets soft branch weights; scores are min-max normalized and combined by weighted-sum fusion (Reciprocal Rank Fusion as comparison), then the top 30 are reranked by a cross-encoder (ms-marco-MiniLM-L-6-v2). Numerical questions get table cell localization; graph context expansion adds the section heading, caption and neighbours; an LLM (gpt-oss-20b) answers only from the numbered evidence with validated citations or says "Not found". Relevance feedback uses session Rocchio query expansion and a capped cross-session memory prior. Served by FastAPI with a React UI that highlights the evidence on the page.

Results (Metrics Used): final pipeline on the TEST split (component = evidence-level retrieval):

- **Recall@1 (component):** 0.488 (dense BGE baseline 0.362)
- **MRR (component):** 0.653 (dense BGE baseline 0.544; paired t-test $p < 0.0001$)
- **nDCG@5 (component):** 0.644
- **Recall@1 / MRR (page level):** 0.733 / 0.828 (random baseline 0.270 / 0.501)
- **ANLS (answer accuracy):** 0.646 final vs 0.486 top-1 baseline (+0.161) and 0.791 oracle
- **Query-type classifier accuracy:** 75.1%
- **Rocchio feedback, residual MRR:** 0.312 vs 0.289 without feedback (+0.023, not significant on TEST)